ReRAM In-Memory SNN for AI Edge Inference

2526_C_0043
NEUROQUAD

TEAM



Nithish Reddy



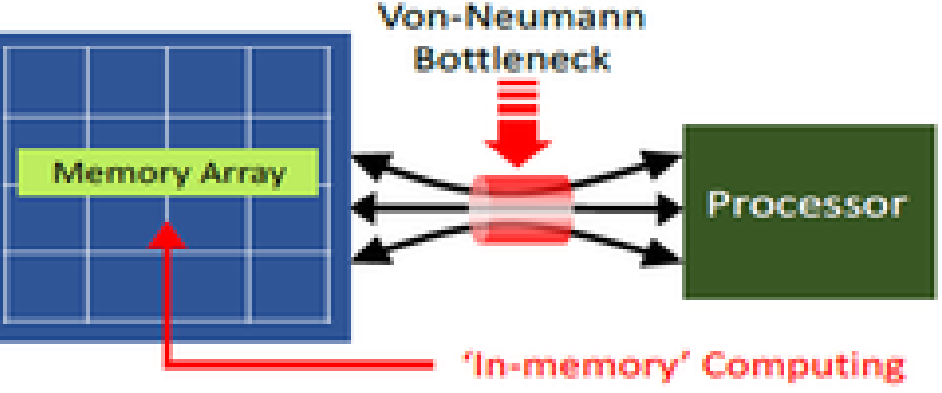
Vishvanath . S



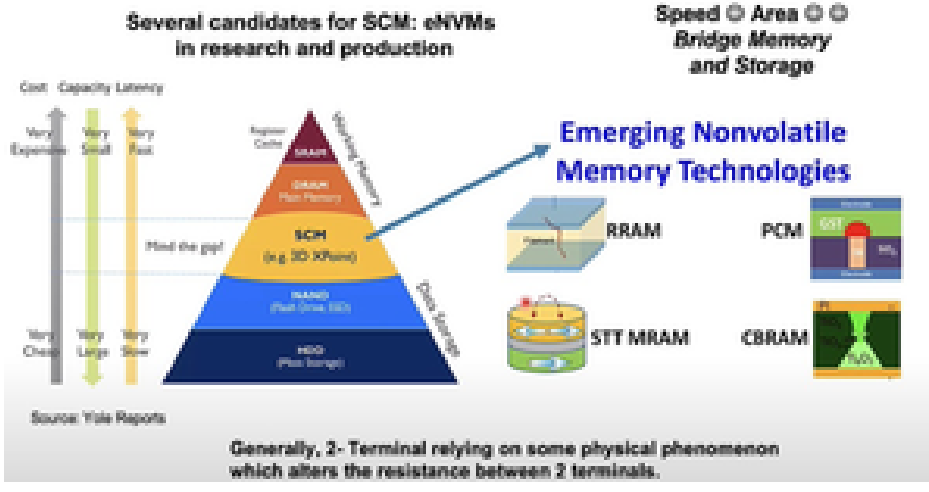
Kavya G · 1st

RESEARCH BACKGROUND

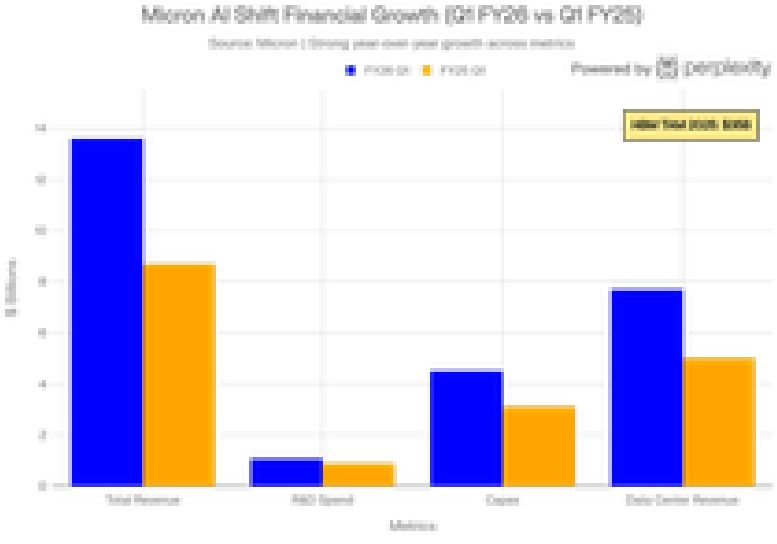
Problem



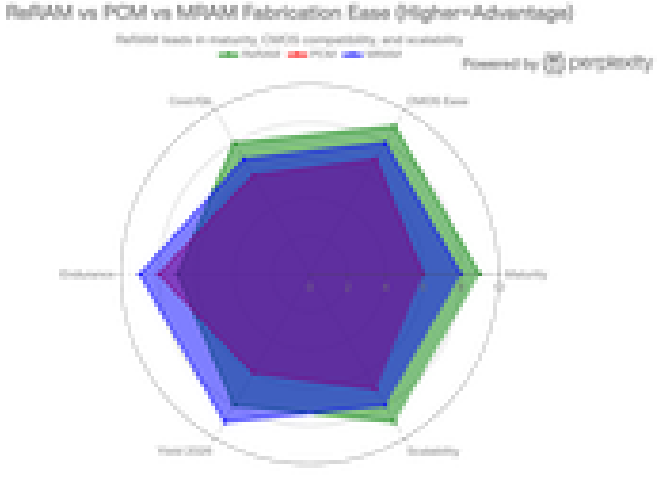
Target



MOTIVATION

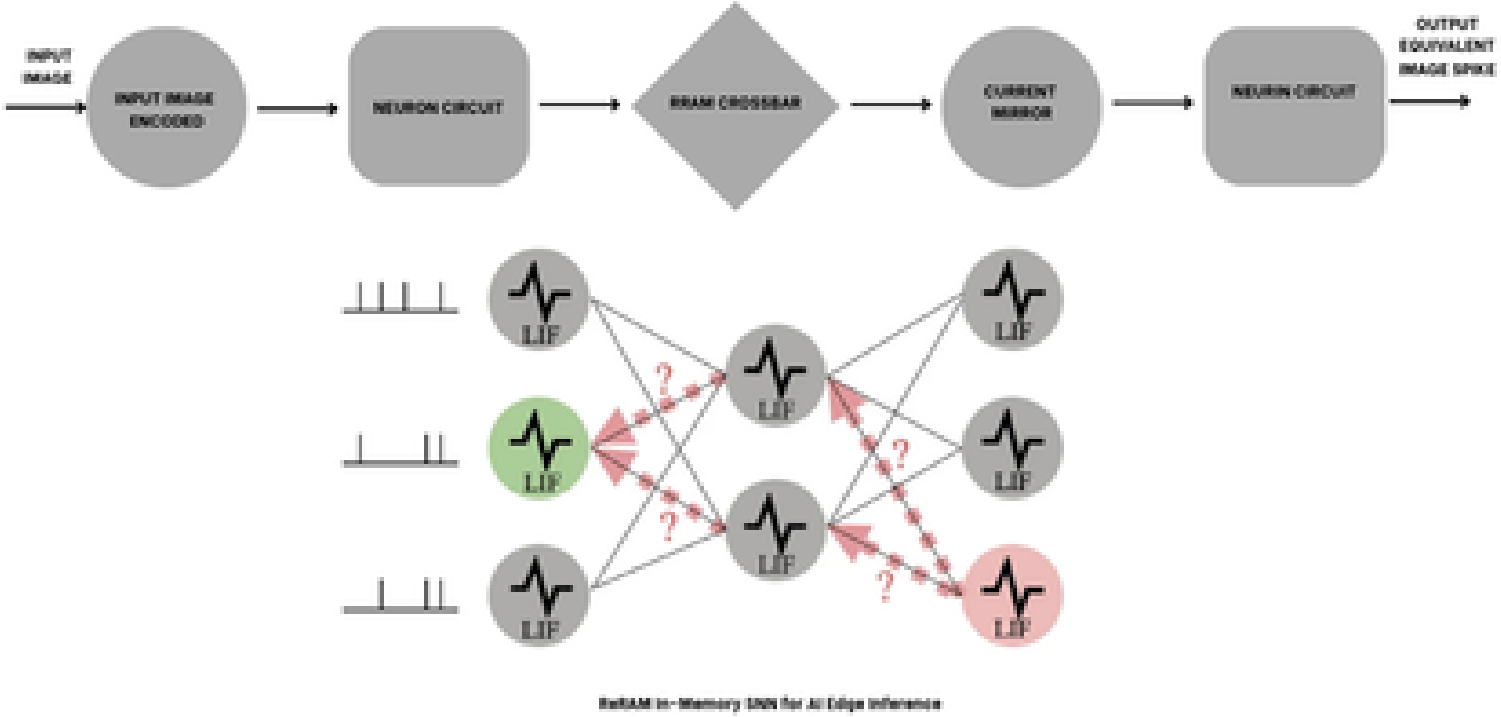


FQ1 2026–MICRON



ReRAM advantages:

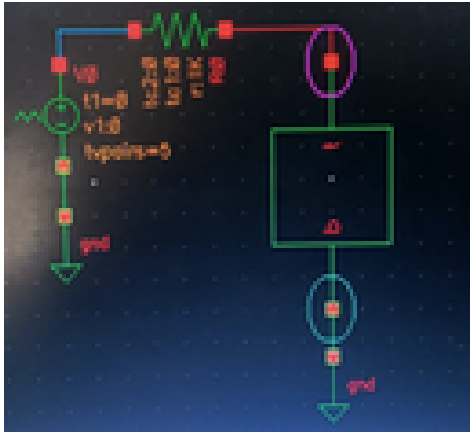
INTRODUCTION

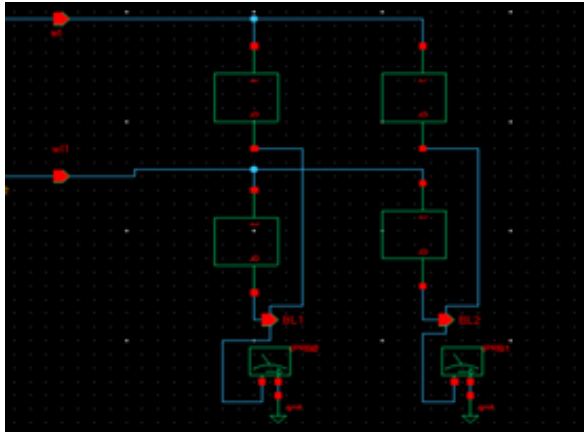


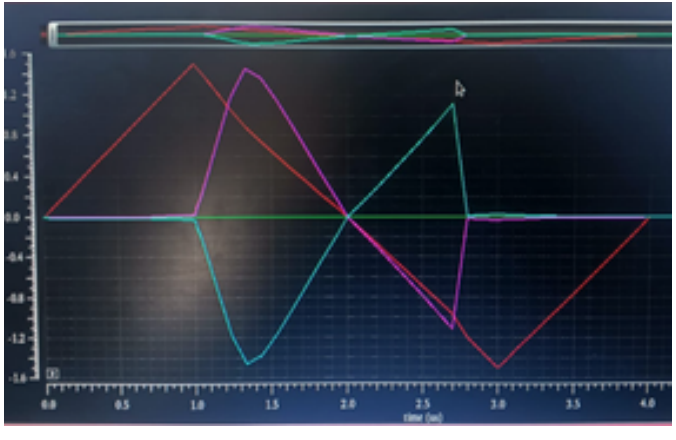
ReRAM in-Memory SNN for AI Edge Inference

FEASABILE MODEL

- **RESISTIVE RANDOM ACCESS MEMORY:**

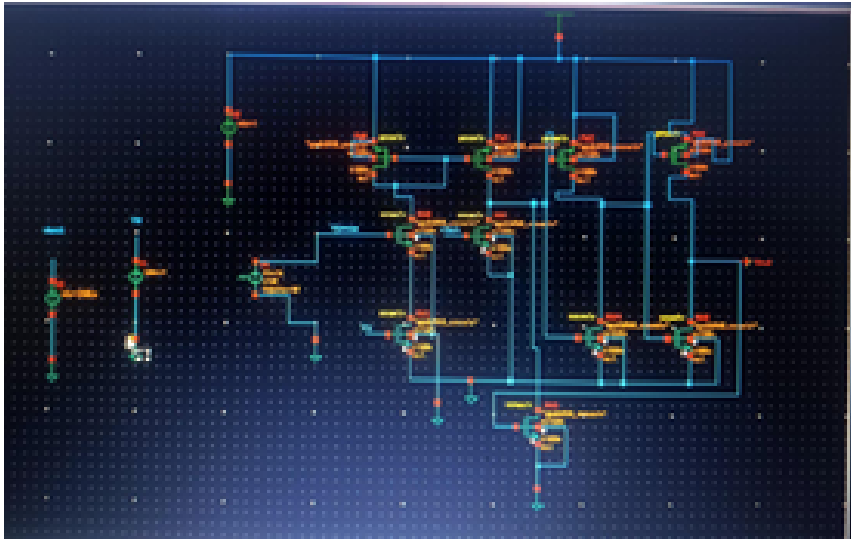


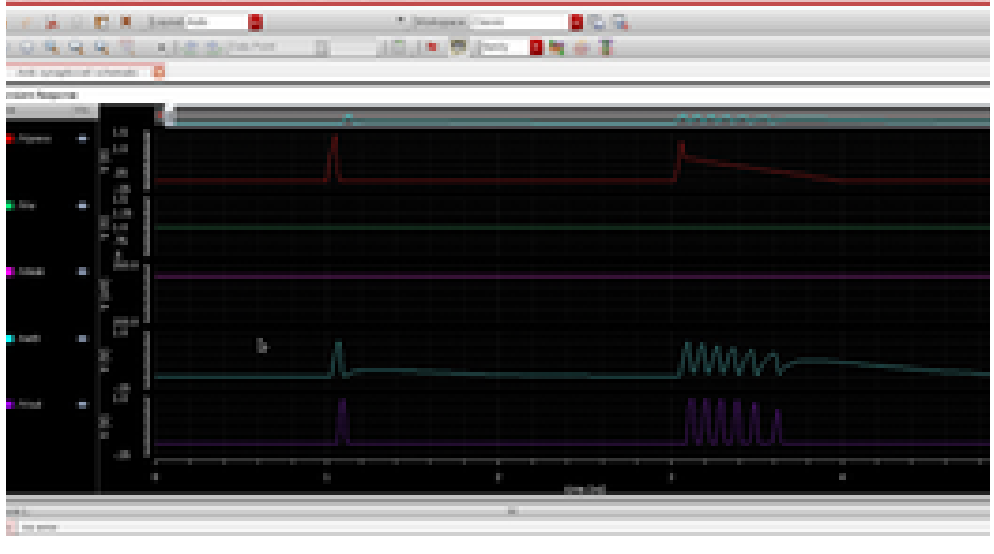




FEASABILE MODEL

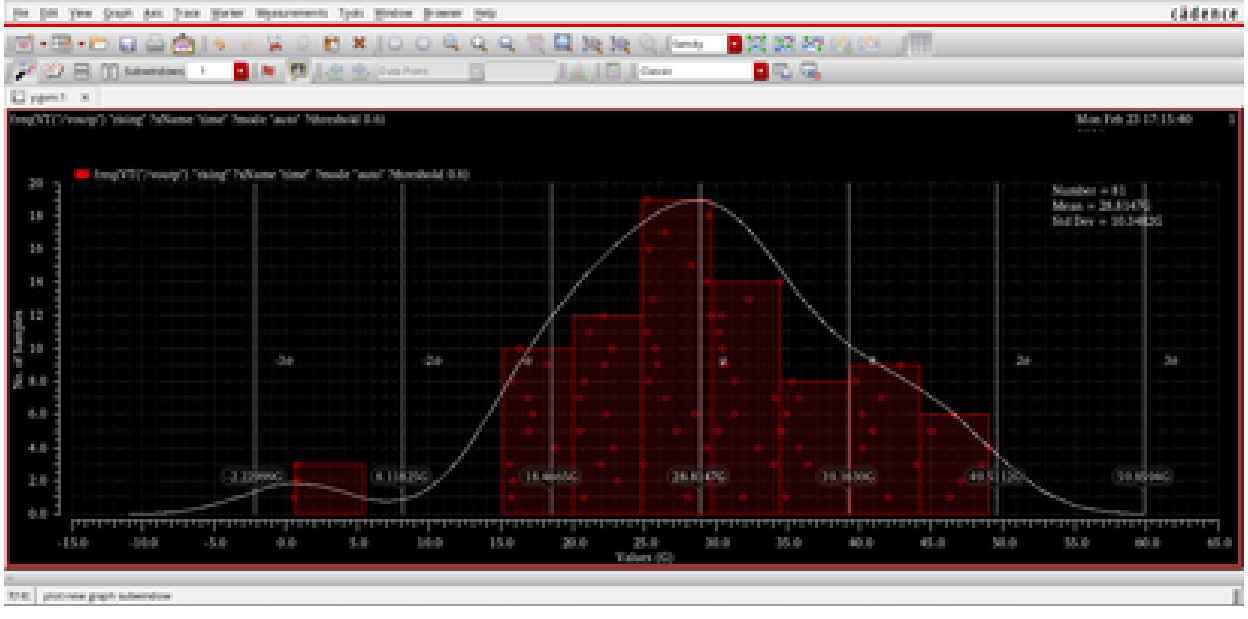
- **SYNAPSTIC RESPONE MODEL (SRM):**





ANALYSES OF VARIANCE

- **SYNAPSTIC RESPONE MODEL (SRM):**



FUTURE WORK

- INTEGRATE FULL CROSSBAR-NEURON ACCELERATOR.
- IMPROVE YIELD (>90%) THROUGH:
- BETTER BIAS STABILIZATION
- LARGER CAPACITANCE OPTIMIZATION
- OFFSET CALIBRATION

TIMELINE

